Lenses

DISCLAIMER

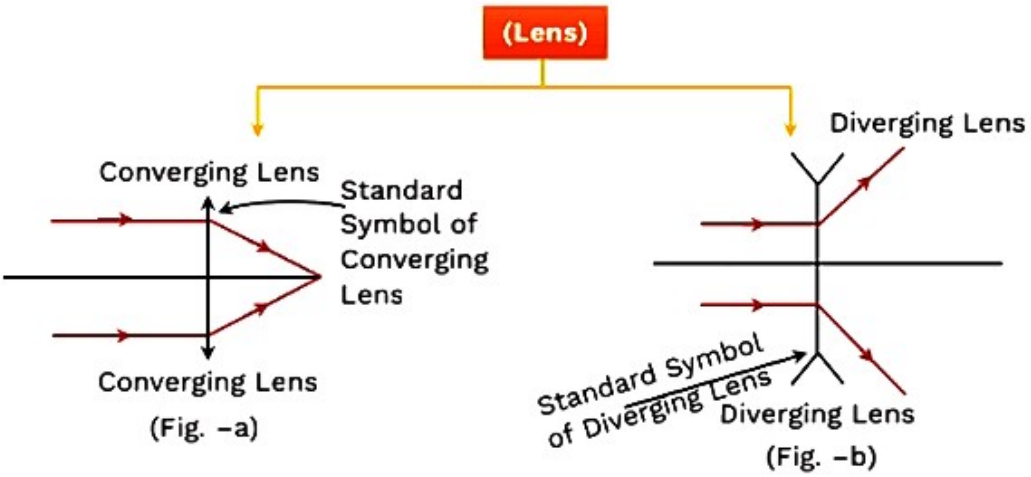
"The content provided herien are created and owned by various authors and licensed to Sorting Hat Technologies Private Limited ("Company"). The Company disclaims all rights and liabilities in relation to the content. The author of the content shall be solely responsible towards, without limitation, any claims, liabilities, damages or suits which may arise with respect to the same."

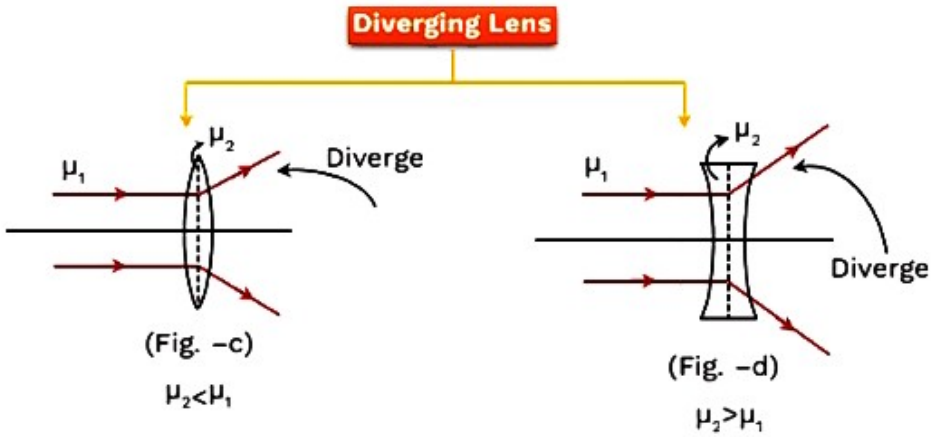
Lenses

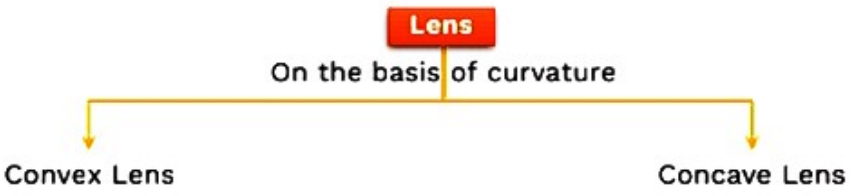
THIN SPHERICAL LENSES

A lens is a piece of transparent material with two refracting surfaces, at least one of them being curved. It may have the surface plane.

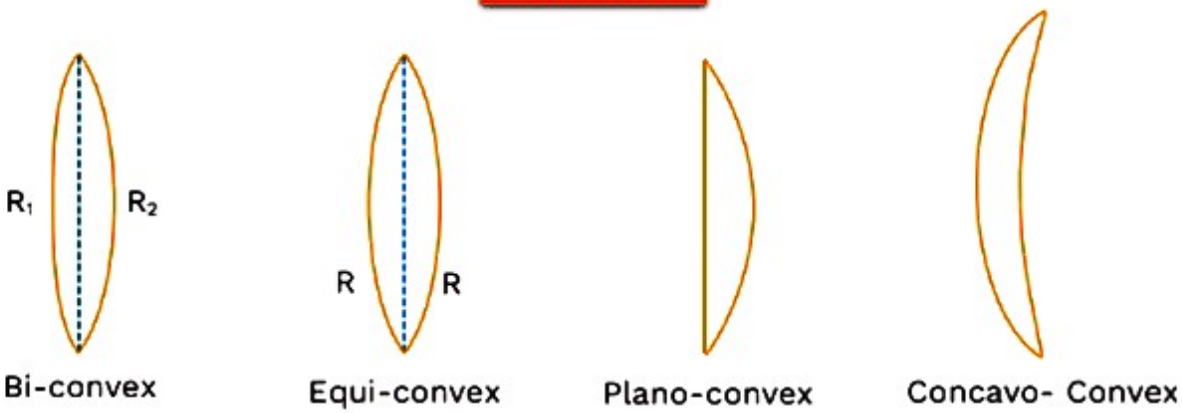
Thin Lens If the thickness of the lens is small (compared to the radius of curvature of spherical surfaces, the object distance, the image distance, etc.). It is said to be thin.



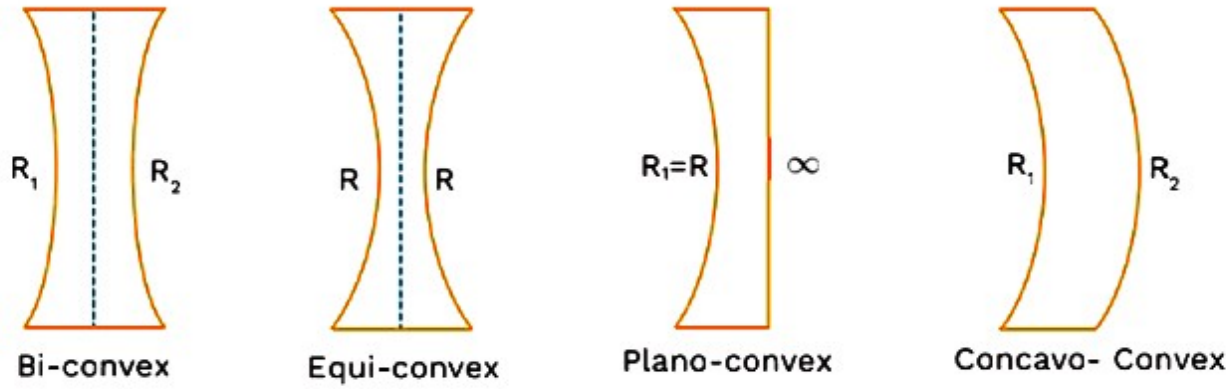




Convex-Lens

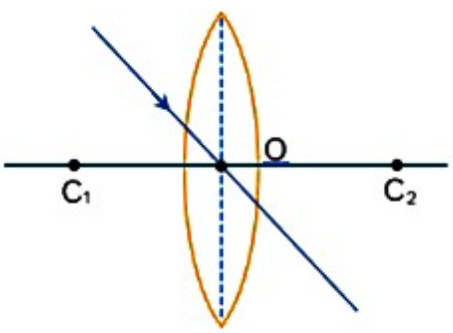


Concave-Lens

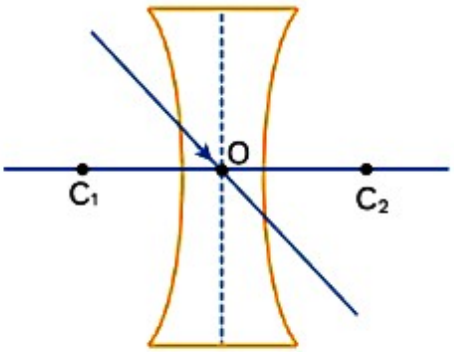


IMPORTANT TERMS AND DEFINITION

Optical Centre optical center O is a point for a given lens through which any ray passes undeviated.

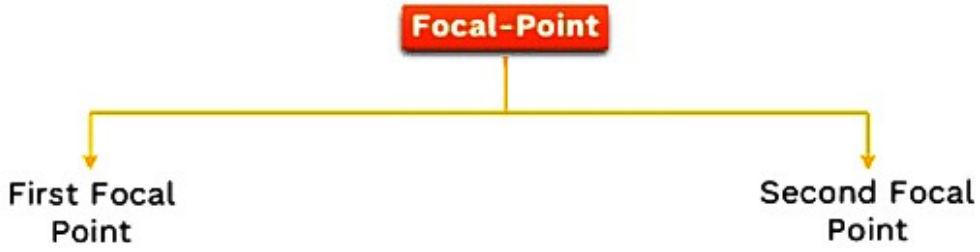


(Fig.-a)



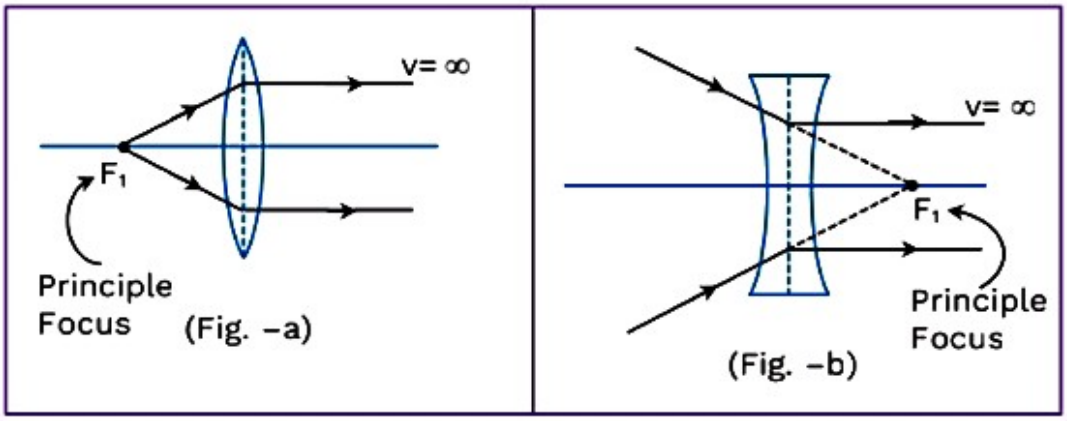
(Fig.-b)

Principal Axis : It is defined as C_1C_2 is a line passing through the optical centre and perpendicular to the lens.



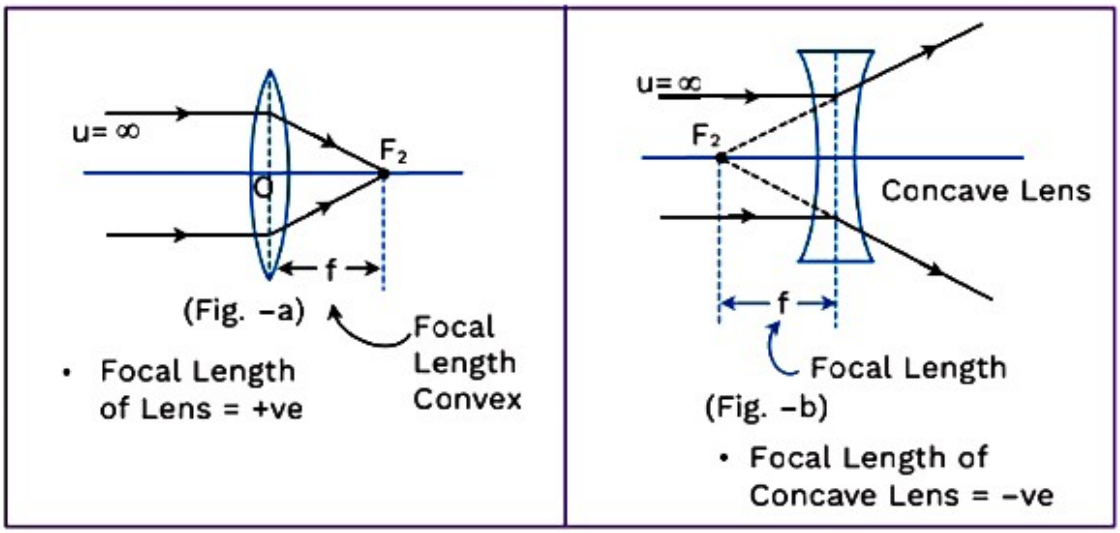
First Focal Point or Principal Focus

The first focal point is an object point on the principal axis for which the image is formed at infinity.



Second Focal Point

The second focal point is on image point on the principal axis for which the object lies at infinity.



Focal Length 'f' : Focal length is defined as the distance between the point where the parallel beam of light appears to converge or converges and the optical center of a lens.

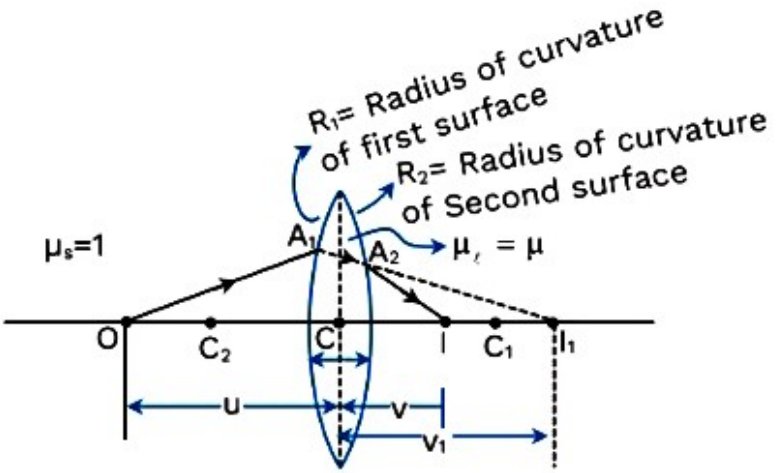
Aperture : Aperture means the effective diameter

$$\text{Intensity} \propto (\text{Aperture})^2$$

Derivation of Lens Makers' Formula

Assumptions Used in The Derivation of Lens Formula

(a) The lens used is thin.
(b) The aperture of the lens is small.
(c) The incident and refracted rays make small angles with the principal axis.
(d) The object is a small object placed on the principal axis.



(Fig.-a)

Consider a thin lens having its optical centre at C and let O be the point object situated on its principal axis, as shown in the figure. Light starting from 'O' strike the first surface of the lens at A_1 and heads I_1, however, refraction takes place at the second surface, thereby giving a final real image at I.

For First-Surface Use Formula for Spherical Refraction

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

μ_2= Refractive index of that medium in which rays it going.
μ_1= Refractive index of that medium from where ray is incoming.
v = Image distance
u = Object Distance.
$\mu_2 = \mu$; $R = R_1$; $u = u$
$\mu_1 = 1$; $v = v_1$

$$\boxed{\frac{\mu}{v_1} - \frac{1}{u} = \frac{\mu - 1}{R_1}} \quad \ldots(1)$$

Now Use Spherical Refraction Formula for Second Surface

$$\boxed{\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}}$$

$\mu_2 = 1$

$\mu_1 = \mu$

$R = R_2$

$u = v_1$

$v = v$

put all these values in above formula

$$\boxed{\frac{1}{v} - \frac{\mu}{v_1} = \frac{1-\mu}{R_2}} \quad \ldots(2)$$

Now add equation (1) and equation (2)

$$\frac{\mu}{v_1} - \frac{1}{u} + \frac{1}{v} - \frac{\mu}{v_1} = \frac{\mu - 1}{R_1} + \frac{1-\mu}{R_2}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{\mu}{R_1} - \frac{1}{R_1} + \frac{1}{R_2} - \frac{\mu}{R_2}$$

$$= \frac{\mu R_2 - R_2 + R_1 - \mu R_1}{R_1 R_2}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{\mu(R_2 - R_1) + (R_1 - R_2)}{R_1 R_2}$$

$$\frac{1}{v} - \frac{1}{u} = \frac{\mu(R_2 - R_1) - (R_2 - R_1)}{R_1 R_2}$$

$$\frac{1}{v} - \frac{1}{u} = (\mu - 1)\left(\frac{R_2 - R_1}{R_1 R_2}\right)$$

$$\frac{1}{v} - \frac{1}{u} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \ldots(3)$$

Now Let Refractive index of medium = μ_s

Refractive index of lens = μ_ℓ

Then lens formula becomes →

$$\frac{\mu_s}{v} - \frac{\mu_s}{u} = (\mu_\ell - \mu_s)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{\mu_\ell}{\mu_s} - \frac{\mu_s}{\mu_s}\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\boxed{\frac{1}{v} - \frac{1}{u} = \left(\frac{\mu_\ell}{\mu_s} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)} \quad \ldots(4)$$

Where,

v = Image distance
u = Object distance
μ_ℓ = Refractive index of lens
μ_s = Refractive index of surrounding
R_1 = Radius of curvature of first surface.
R_2 = Radius of curvature of 2nd surface.

NOTE

(1) Use u, v, R_1, and R_2 with the proper sign.

- If Object placed at infinity; image formed at 'f'.
  $u = \infty$; $v = f$

$$\frac{1}{v} - \frac{1}{u} = \left(\frac{\mu_\ell}{\mu_s} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} - \frac{1}{\infty} = \left(\frac{\mu_\ell}{\mu_s} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) \quad \ldots(5)$$

Now compare equation (4) and equation (5)

$$\boxed{\frac{1}{v} - \frac{1}{u} = \frac{1}{f}}$$

Lens formula

For air $(\mu_s = 1; \mu_\ell = u)$

$$\boxed{\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)}$$

Lens maker formula.

IMPORTANT POINT REGARDING LENS MAKER FORMULA

(a) Lens must be thin.

(b) Rays must be paraxial

(c) $R_1 = R_1$ is the radius of curvature of that surface at which ray incident first.

(d) $R_2 = R_2$ is the radius of curvature that surface at which rays strike secondly.

SUMMARY

(1) $$\boxed{\frac{1}{v}-\frac{1}{u}=\left(\frac{\mu_\ell}{\mu_s}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)}$$

If $\mu_s = 1$ (air) ; $\mu_\ell = \mu$

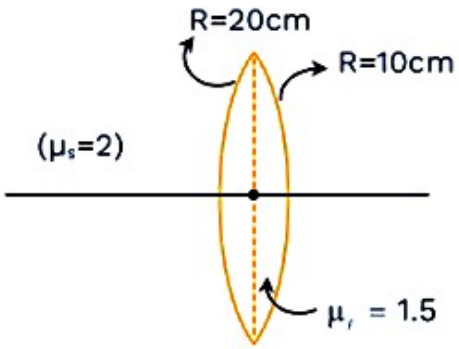
(2) $$\boxed{\frac{1}{v}-\frac{1}{u}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)}$$

(3) $u = \infty$; $v = f$

$$\boxed{\frac{1}{f}=(\mu-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)}$$

(4) $$\boxed{\frac{1}{v}-\frac{1}{u}=\frac{1}{f}}$$

NOTE → All these formulae are valid for convex as well as a concave lens.

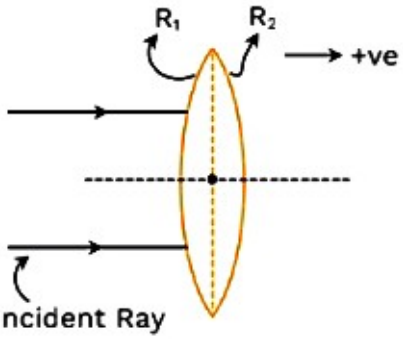


Find the focal length of the lens; All required data are given in the figure. Also, draw a ray diagram.

Sol: Use Lens maker formula

Case -1 When rays strike from left to right.

$$\boxed{\frac{1}{f}=\left(\frac{\mu_\ell}{\mu_s}-1\right)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)}$$



R_1 = +20 (Since rays strike on it first)

R_1 = –10 cm (Since distance measured opposite direction of incident ray)

$$\frac{1}{f}=\left(\frac{1.5}{2}-1\right)\left(\frac{1}{+20}-\frac{1}{-10}\right)$$

$$\frac{1}{f}=\left(\frac{1.5-2}{2}\right)\left(\frac{1}{20}+\frac{1}{10}\right)$$

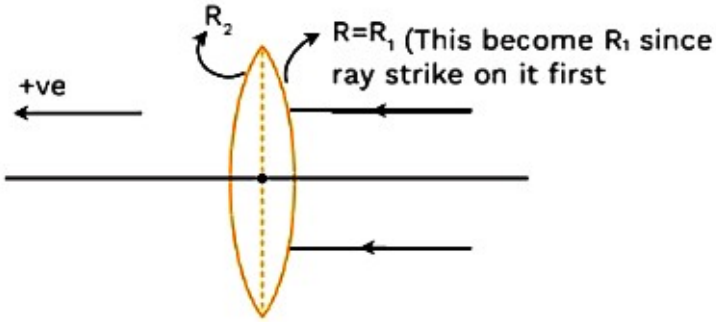
$$=\frac{-0.5}{2}\left(\frac{1+2}{20}\right)$$

$$\frac{1}{f}=\frac{-0.5\times 3}{40}$$

$$f=-\frac{40}{0.5\times 3}\Rightarrow \boxed{f=-\frac{80}{3}\text{cm}}$$

Case -2

When rays strike from right to left.



$$\left.\begin{aligned} R_1 &= +10\,cm \\ R_2 &= -20\,cm \\ \mu_s &= 2 \\ \mu_\ell &= 1.5 \end{aligned}\right\} \Rightarrow \quad \frac{1}{f} = \left(\frac{\mu_\ell}{\mu_s} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f} = \left(\frac{1.5}{2} - 1\right)\left(\frac{1}{10} - \frac{1}{-20}\right)$$

$$\frac{1}{f} = \left(\frac{1.5-2}{2}\right)\left(\frac{2+1}{20}\right)$$

$$\frac{1}{f} = \frac{-0.5}{2}\left(\frac{3}{20}\right)$$

$$f = -\frac{40}{0.5 \times 3} \Rightarrow \boxed{f = \frac{-80}{3}\,cm}$$

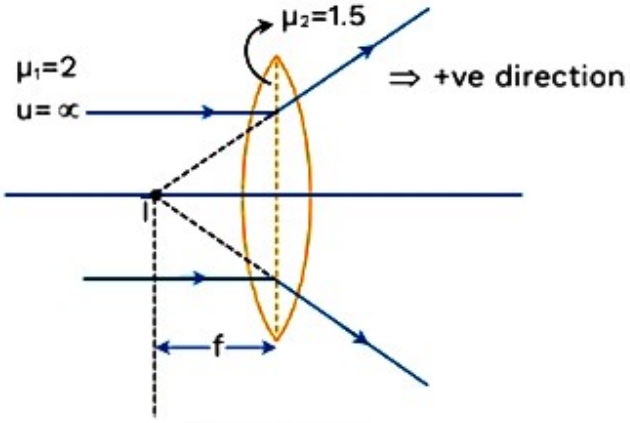
Conclusion

The focal length of convex may be positive or may be negative depends upon the refractive index of the medium on the lens.

The focal length of the convex lens may be positive or may be negative.

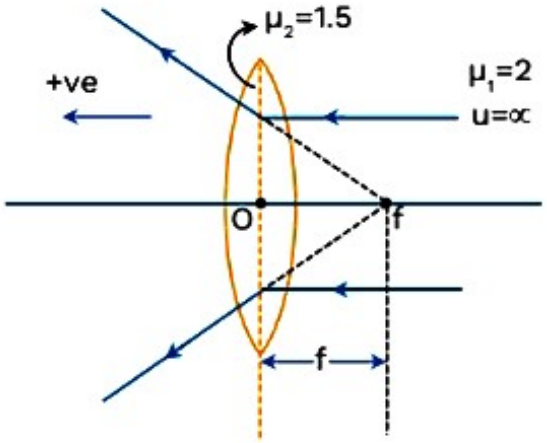
The direction of the incident ray does not affect the focal length of a lens.

Case -1 Ray Diagram



f=−80/3 cm

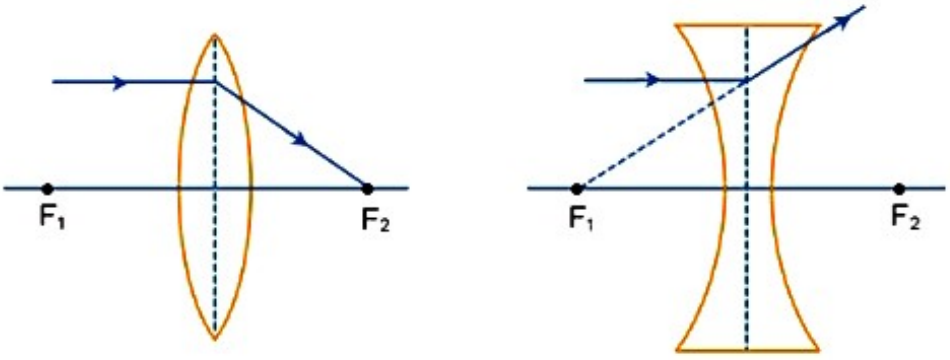
Case -2 Ray Diagram



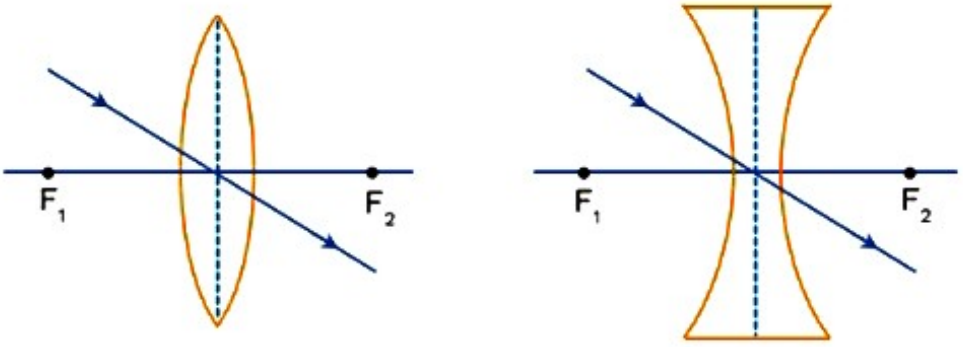
f=−80/3 cm

Rules for Obtaining Images in Lenses:

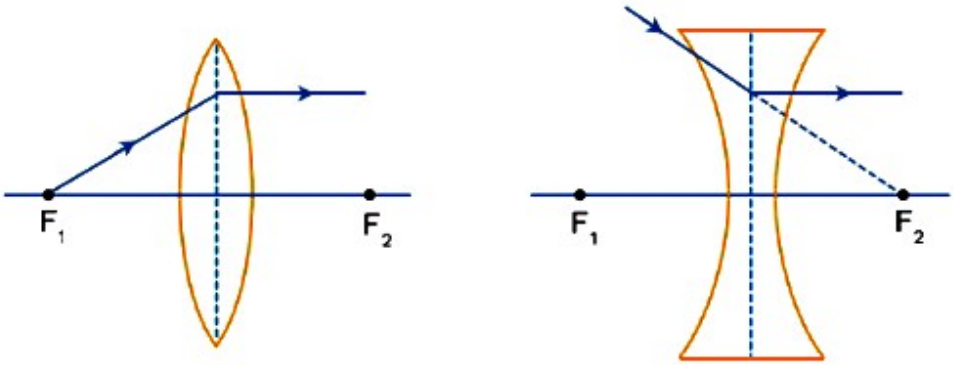
1. When a ray of light parallel to the principal axis, after reaction through the lens, converges to the focus (in case of a converging lens) or appears to diverge from the focus (in case of a diverging lens).



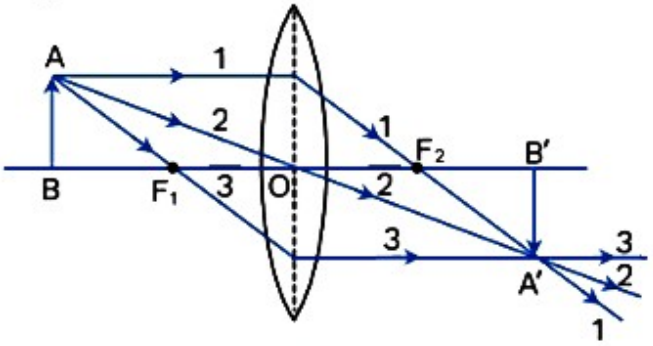
2. A ray passing through the optical center goes through the lens undeviated.



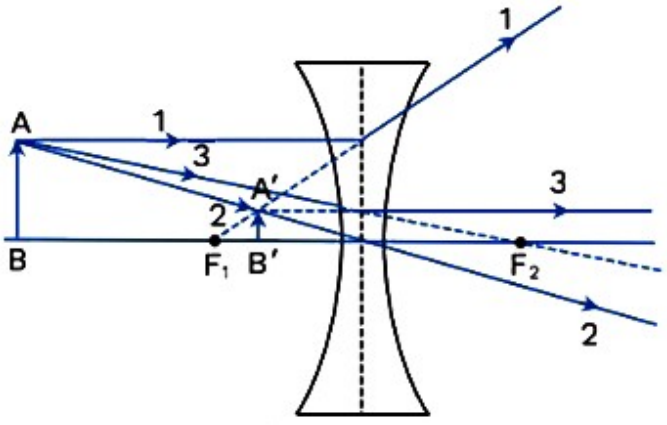
3. A ray passing through the focus (in case of a convex lens) or appearing to pass through the focus (in case of a concave lens) becomes parallel to the principal axis after refraction through the lens.



Any two of the above rays can be used to obtain the location of the image.

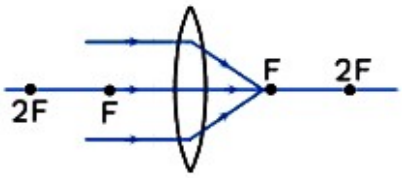
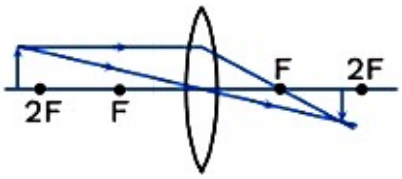
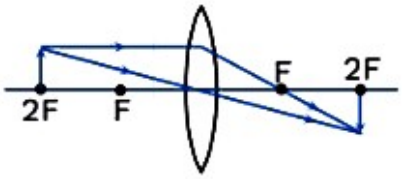
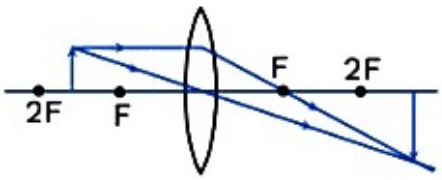
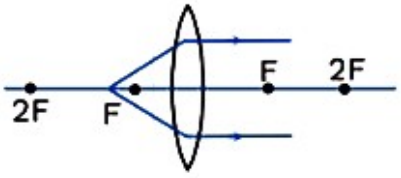
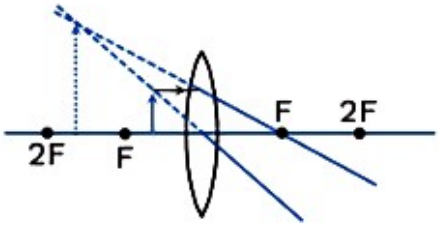


Convex Lens



Concave Lens

Image Formation by Convex Lens:

OBJECT POSITION	DIAGRAM	POSITION OF IMAGE	NATURE AND SIZE OF IMAGE
At infinity		At the principal Focus (F) or in the focus plane	Real, inverted and extremely diminished
Beyond 2F		Between F and 2F	Real, inverted and diminished
At 2F		At 2F	Real, inverted and of same size as the object
Between F and 2F		Beyond 2F	Real, inverted and magnified
At F		At infinity	Real, inverted and highly magnified
Between F and Optical centre		On the same side as the object	Virtual, erect and magnified

Q **Calculate the distance of an object from a convex lens of the focal length of 10 cm if the image formed is two times the size of an object. It is given that the Focal length of the lens is 10 cm.**

Sol: A convex lens form both type of images, real as well as virtual. Since, nature of the image is not mentioned here, so we will have to consider both the case.

Case I When image is real

In this case v is positive and u is negative with $|v| = 2|u|$, so if

$u = -x$ then $v = 2x$ and $f = 10$ cm

Substituting in $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{2x} + \frac{1}{x} = \frac{1}{10}$$

$$\Rightarrow \quad \frac{3}{2x} = \frac{1}{10}$$

$$\Rightarrow \quad x = 15 \text{ cm}$$

$x = 15$ cm, means object les between F and 2F.

Case II When image is virtual

In this case v and u both are negative. So if

$u = -y$ the $v = -2y$ and $f = 10$ cm

Substituting in, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$, we get

$$\Rightarrow \quad \frac{1}{-2y} + \frac{1}{y} = \frac{1}{10}$$

$$\Rightarrow \quad \frac{1}{2y} = \frac{1}{10}$$

$$\Rightarrow \quad y = 5 \text{ cm}$$

$y = 5$ cm, means object lies between F and P.

Magnification Concept in Lenses:

1. **Linear Magnification (m):** Linear, lateral or transverse Magnification (m) produced by a lens; we can define it as the ratio of the height of an image to the height of the object. Mathematically,

$$m = \frac{\text{Height of Image}}{\text{Height of Object}} = \frac{I}{O}$$

Since triangles ABC and A'B'C are similar, so

$$\frac{A'B'}{BA} = \frac{CA'}{CA}$$

Using Conventions, we get

$A'B' = -I$

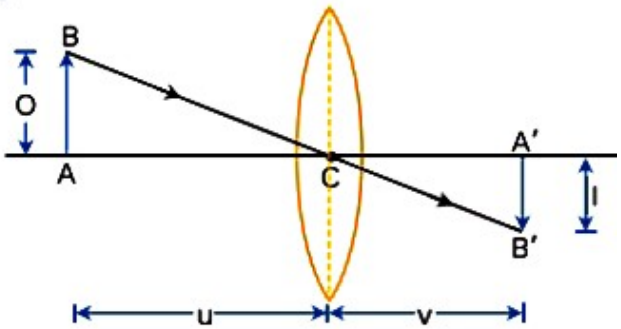
$BA = O$

$CA = -u$

$CA' = +v$

$$\Rightarrow \quad \frac{I}{O} = \frac{v}{u}$$

$$\Rightarrow \quad \boxed{m = \frac{I}{O} = \frac{v}{u}}$$



Please note that for both the lens and mirror we have

m_{real} = Negative i.e. $m_{real} < 0$

$m_{virtual}$ = Positive i.e. $m_{virtual} > 0$

2. **Axial magnification** Axial magnification is the ratio of the size of an image along the principal axis to the size of the object along the principal axis. So

$$\boxed{m_{axial} = \frac{\text{Size of Image along Principal Axis}}{\text{Size of Object along Principal Axis}} = \frac{dv}{du}}$$

$$\Rightarrow \quad m_{axial} = \frac{dv}{du} = \frac{v^2}{u^2} = m^2$$

3. **Areal Magnification** Areal magnification is the ratio of the area of the image to the area of the object.

$$m_{areal} = \frac{\text{Area of Image}}{\text{Area of Object}} = \frac{A_I}{A_O}$$

$$m_{areal} = \frac{A_I}{A_O} = \frac{v^2}{u^2} = m^2$$

Q **Suppose a thin convex lens of focal length 0.3 m and an object is approaching toward the lens with a speed of 0.01 ms^{-1}. Calculate the magnitudes of the rates of change of position and also the magnitudes of the rates of change lateral magnification of image when the object is at a distance of 0.4m from the lens.**

Sol: Differentiating the lens formula $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ with respect to time, we get

$$-\frac{1}{v^2}\frac{dv}{dt}+\frac{1}{u^2}\frac{du}{dt}=0 \qquad \{\because f=\text{constant}\}$$

$$\Rightarrow \left(\frac{dv}{dt}\right)=\left(\frac{v^2}{u^2}\right)\frac{du}{dt} \qquad ...(1)$$

Further, by substituting proper values in the lens formula, we get

$$\frac{1}{v}+\frac{1}{0.4}=\frac{1}{0.3} \qquad \{\because u=-0.4\,m, f=0.3\,m\}$$

$\Rightarrow$ v = 1.2 m

Put these values in equation 1 ; we get

The magnitude of the rate of change of position of the image is

$$\frac{dv}{dt}=0.09\,ms^{-1}$$

Lateral magnification, $m=\frac{v}{u}$

The rate of change of lateral magnification is given by

$$\frac{dm}{dt}=\frac{u\frac{dv}{dt}-v\frac{du}{dt}}{u^2}=\frac{(-0.4)(0.09)-(1.2)(0.01)}{(0.4)^2}$$

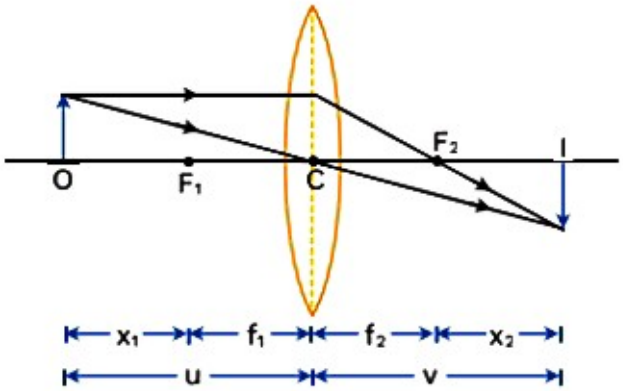
$$\Rightarrow \frac{dm}{dt}=-0.3 \text{ per second}$$

$\Rightarrow$ So Magnitude of the rate of change of lateral magnification is $\frac{dm}{dt}=0.3$ per second.

NEWTON'S FORMULA

When the object and the image distances are measured from the first and second principal foci instead of measuring them from the optical center, and if x_1 is the distance of the object from the first focus, x_2 is the distance of the image from the second focus. For the focal length of the lens to be f, we have

$$u=-(f+x_1),\ v=f+x_2$$



According to the lens formula, we have

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{1}{f + x_2} - \frac{1}{-(f + x_1)} = \frac{1}{f} \Rightarrow \frac{f + x_1 + f + x_2}{(f + x_2)(f + x_1)} = \frac{1}{f} \Rightarrow f^2 = x_1 x_2$$

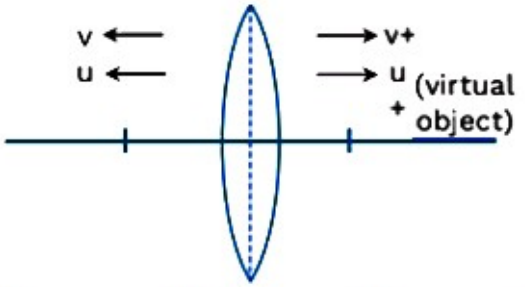
$$\Rightarrow \quad \boxed{f = \sqrt{x_1 x_2}}$$

This is called Newton's formula.

where $x_1 = F_1O$ = distance of object form first principal focus F_1

$x_2 = F_2I$ = distance of image from second principal focus F_2

u–v DIAGRAM FOR CONVEX LENS

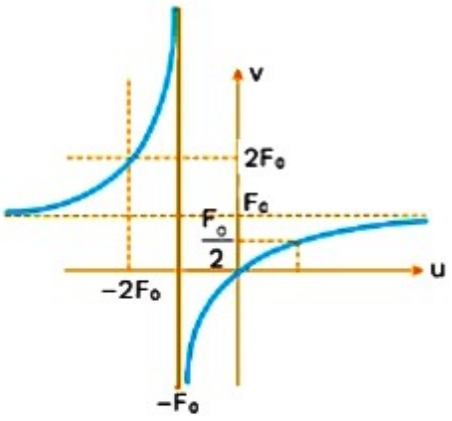


Convex Lens $\quad v = \frac{uf}{u + f}$,

f is positive let $f = F_0$

$$v = \frac{uF_0}{u + F_0}$$

u	$-\infty$	$-2F_0$	$-F_0$	0	F_0	∞
v	F_0	$2F_0$	$\pm\infty$	0	$\frac{F_0}{2}$	F_0



"u–v graph of convex lens"

DISPLACEMENT METHOD

Consider an object and a screen fixed at a distance D apart. Let a lens of focal length f be placed between the object and the screen.

From figure we observe that

$u + v = D \Rightarrow v = D - u$

Also from Lens formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

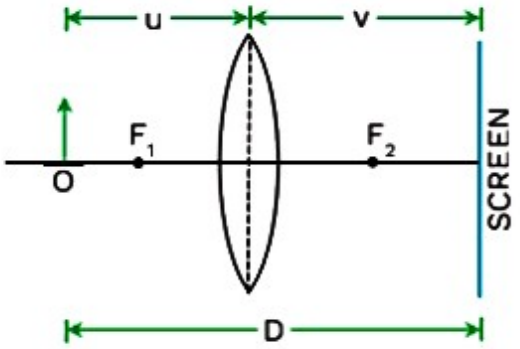
$$\Rightarrow \frac{1}{D-u} - \frac{1}{-u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{D-u} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow Df = (D - u)u$$

$$\Rightarrow u^2 - Du + Df = 0$$

$$\Rightarrow u = \frac{D \pm \sqrt{D^2 - 4fD}}{2}$$



For u to be mathematically real,

$D^2 - 4fD \geq 0$

$\Rightarrow D \geq 4f$

PROBLEM SOLVING TRICK(S)

So, if the object and the screen are placed at a distance less than 4f, then a virtual image will be formed. Hence, for a real image to be formed $D \geq 4f$.

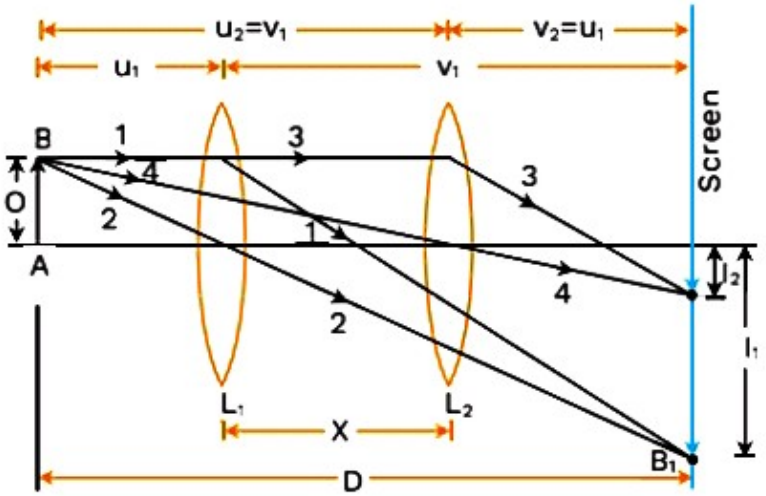
Case -I For D = 4f

$$u = v = \frac{D}{2}$$

i.e., the lens is placed exactly between the object and the screen.

Case -II For D > 4f

We get two different positions of the lens (L_1 and L_2) for which the image of an object on the screen is distinct and clear.



L_1 First Position of Lens
L_2 Second position of the Same Lens (shown in figure)

NOTE

Do not Develop a misconception that there are two lenses; in fact, the same lens is displaced through x from position L_1 to L_2.

The object distance for these two positions are given by

$$u_1 = \frac{D - \sqrt{D^2 - 4fD}}{2} \quad ...(1)$$

$$u_2 = \frac{D + \sqrt{D^2 - 4fD}}{2} \quad ...(2)$$

Since u + v = D, so

$$v_1 = \frac{D + \sqrt{D^2 - 4fD}}{2} \quad ...(3)$$

$$v_2 = \frac{D - \sqrt{D^2 - 4fD}}{2} \quad ...(4)$$

We observe that

$$u_1 = v_2 = u \text{ (say)} \quad ...(5)$$

$$v_1 = u_2 = v \text{ (say)} \quad ...(6)$$

Let the lens be displaced through x, then we observe from figure that

$$x = v_1 - u_1 = \sqrt{D^2 - 4fD} \quad ...(7)$$

$$\Rightarrow \quad x^2 = D^2 - 4fD$$

$$\Rightarrow \quad \boxed{f = \frac{D^2 - x^2}{4D}} \quad ...(8)$$

Using (7) in (1), (2), (3) and (4), we get

$$u_1 = v_2 = \frac{D - x}{2} = u \quad ...(9)$$

$$v_1 = u_2 = \frac{D + x}{2} = v \quad ...(10)$$

If m_1 is the magnification for the first position of lens i.e., L_1, then

$$m_1 = \frac{I_1}{O} = \frac{v_1}{u_1} = \frac{v}{u} = \frac{D + x}{D - x} \quad ...(11)$$

If m_2 is the magnification for the second position of the lens i.e. L_2, then

$$m_2 = \frac{I_2}{O} = \frac{v_2}{u_2} = \frac{u_1}{v_1} = \frac{u}{v} = \frac{D - x}{D + x} \quad ...(12)$$

From (11) and (12), we observe that

$$m_1 m_2 = 1 \quad ...(13)$$

So, if magnification for position L_1, is m, then magnification for position L_2 is $\frac{1}{m}$.

Also, from (13), we get

$$\frac{I_1}{O}\frac{I_2}{O} = 1$$

$$\Rightarrow \quad O^2 = I_1 I_2$$

$$\boxed{O = \sqrt{I_1 I_2}}$$

i.e., the size of the object (O) is the geometric mean of the sizes of the image for two positions of lens L_1 and L_2.

Also,

$$m_1 - m_2 = \frac{D+x}{D-x} - \frac{D-x}{D+x}$$

$$\Rightarrow \quad m_1 - m_2 = \frac{4Dx}{D^2 - x^2}$$

$$\Rightarrow \quad m_1 - m_2 = \frac{x}{\left(\frac{D^2 - x^2}{4D}\right)}$$

$$\Rightarrow \quad m_1 - m_2 = \frac{x}{f}$$

$$\Rightarrow \quad f = \frac{x}{m_1 - m_2} \qquad \ldots(15)$$

Further if $m_1 = m$, then $m_2 = \frac{1}{m}$

$$\Rightarrow \quad f = \frac{mx}{m^2 - 1} \qquad \ldots(16)$$

Finally, we observe that $\boxed{\frac{m_1}{m_2} = \left(\frac{D+x}{D-x}\right)^2}$

Q **Calculate the focal length of a lens with positive optical power for which two real images are formed for two different positions of the lens between an object and a screen kept 96 cm apart from each other? Given the ratio of length of the two images as 4.**

Sol: Since, $\frac{m_1}{m_2} = 4$

$$\Rightarrow \quad \left(\frac{D+x}{D-x}\right)^2 = 4$$

$$\Rightarrow \quad \frac{D+x}{D-x} = 2$$

Substituting D = 96 cm, we get

$$\frac{96+x}{D-x} = 2$$

$$\Rightarrow \quad x = 32 \text{ cm}$$

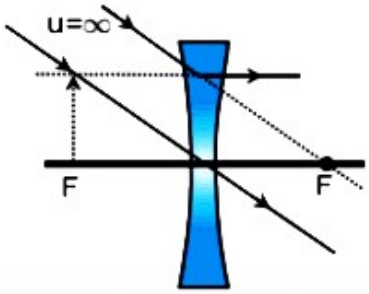
Since, $f = \frac{D^2 - x^2}{4D}$

$$\Rightarrow \quad f = 21.33 \text{ cm}$$

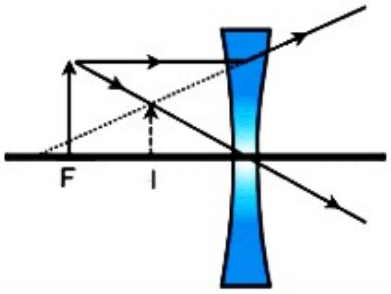
IMAGE FORMATION FOR CONCAVE LENS (DIVERGENT LENS)

Image is virtual, diminished, erect, towards the object, m = +ve

(i) object is placed at infinity
Image
At F virtual erected
diminished (m << + 1)



(ii) Object is placed in front of lens
Image
between F and optical centre
virtual erect diminished (m < + 1)



Sign shortcut for object/image for lens

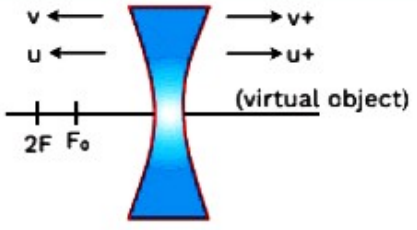
Real object	u must be negative
Real image	v must be positive
Virtual object	u must be positive
Virtual image	v – must be negative

u–v DIAGRAM FOR CONCAVE LENS

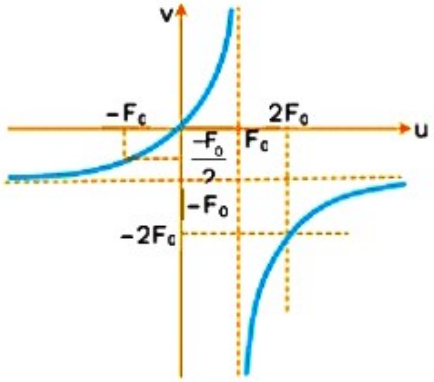
Concave lens $v = \frac{uf}{u+f}$

f is negative, let $f = -F_0$

$$v = \frac{-uF_0}{u+F_0}$$



u	$-\infty$	$-F_0$	0	F_0	$2F_0$
v	F_0	$\frac{-F_0}{2}$	0	$\pm\infty$	$-2F_0$



RELATION BETWEEN OBJECT AND IMAGE VELOCITY

Case (I) - When the object is moving along the principal axis -

Let a point object is moving along the principal axis of a lens of focal length f. We can relate the object and image positions by the lens formula $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$.

Differentiating mirror formula on both sides w.r.t. time we get, $-\frac{1}{v^2}\frac{dv}{dt}+\frac{1}{u^2}\frac{du}{dt}=0$

(Since focal length of the lens remains constant)

$$\boxed{\frac{dv}{dt}=+\frac{v^2}{u^2}\frac{du}{dt}\Rightarrow (V_{IL})_{\parallel}=\left(\frac{v^2}{u^2}\right)(V_{oL})_{\parallel}} \quad ...(1)$$

$(V_{IL})_{\parallel}$ = Velocity of image w.r.t. lens along the principal axis

$(V_{OL})_{\parallel}$ = Velocity of object w.r.t. lens along the principal axis.

Important Points

Image speed = m^2 × Object speed

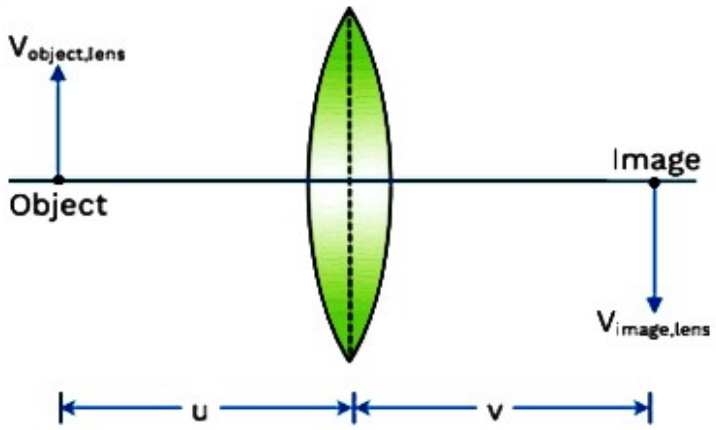
In vector notation, Eq.1 can be written as

$$\boxed{\bar{V}_{image,lens}=m^2\bar{V}_{object,lens}\Rightarrow(\bar{V}_{image}-\bar{V}_{lens})=m^2(\bar{V}_{object}-\bar{V}_{lens})}$$

Here it should be noted that the direction of the velocity of object and image w.r.t. lens is the same to each other.

Case (ii). When an object is moving normal to the principal axis

In this case, a point object is moving normal to the principal axis of a concave mirror, as shown in the figure. In this situation the object distance will not change, it makes image distance also constant.



For constant values of v and u, the value of the transverse magnification is also constant or $m = \frac{I}{O} = \frac{v}{u}$ is a constant.

We can write, $I = mo$. ...(i)

Differentiate Eq. (i) with respect to time, we get

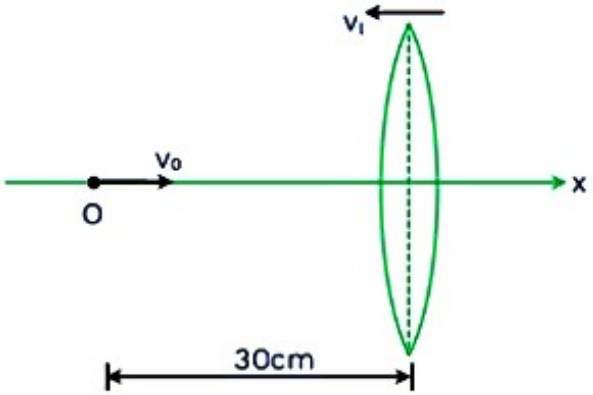
$$\frac{dI}{dt} = m\frac{dO}{dt}$$

$$\Rightarrow \left(V_{image,lens}\right)_y = m\left(V_{object,lens}\right)_y$$

Here $\frac{dI}{dt} = \left(V_{image,lens}\right)_y$ = velocity of image w.r.t. lens in the direction perpendicular to the principal axis,

and $\frac{dO}{dt} = \left(V_{object,lens}\right)_y$ = velocity of object w.r.t. lens in the direction perpendicular to the principal axis.

Q **A point object has velocity v_0 = 2cm/s when it is placed at a distance of 30 cm from a lens moving with a speed v_{lens} = 2 cm/s towards the left. If the focal length of the lens is f = 20 cm, find the velocity of the image.**



Sol: The Velocity of image w.r.t. lens along the principal axis is given by

$$\bar{V}_{image,lens} = m^2 \bar{V}_{object,lens}$$

$$\Rightarrow \quad \left(\bar{V}_{image} - \bar{V}_{lens}\right) = \left(\frac{v}{u}\right)^2 \left(\bar{V}_{object} - \bar{V}_{lens}\right) \qquad ...(i)$$

Using $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} - \frac{1}{(-30)} = \frac{1}{(+20)} \Rightarrow v = +60\,cm$

Substituting, u = −30cm, v = +60 cm, $\bar{V}_{object} = 2\hat{i}, \bar{V}_{lens} = -2\hat{i}$, is Eq. (i)

$$\bar{V}_{image} = \left(\frac{+60}{-30}\right)^2 [(2\hat{i}) - (-2\hat{i})] + (-2\hat{i}) = 14\hat{i}\,cm/s$$

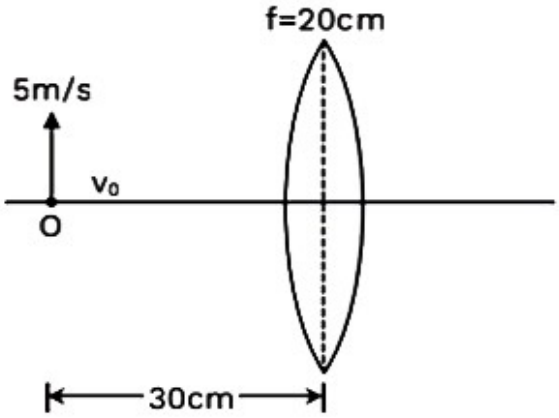
Hence image will move with a velocity of 14 cm/s towards the right.

Q **A point object has velocity v_0 = 5 cm/s perpendicular to the principal axis of a fixed lens. The object is kept at a distance of 30 cm from the lens of focal length f = 20 cm, find the velocity of the image.**

Sol: The velocity of the image with respect to lens perpendicular to the principal axis

$$\boxed{\left(V_{image,lens}\right)_y = \left(\frac{v}{u}\right)\left(V_{object,lens}\right)_y} \qquad ...(1)$$

Apply lens equation $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$



$$\frac{1}{v}-\frac{1}{-30}=\frac{1}{20} \Rightarrow v = +60\,cm$$

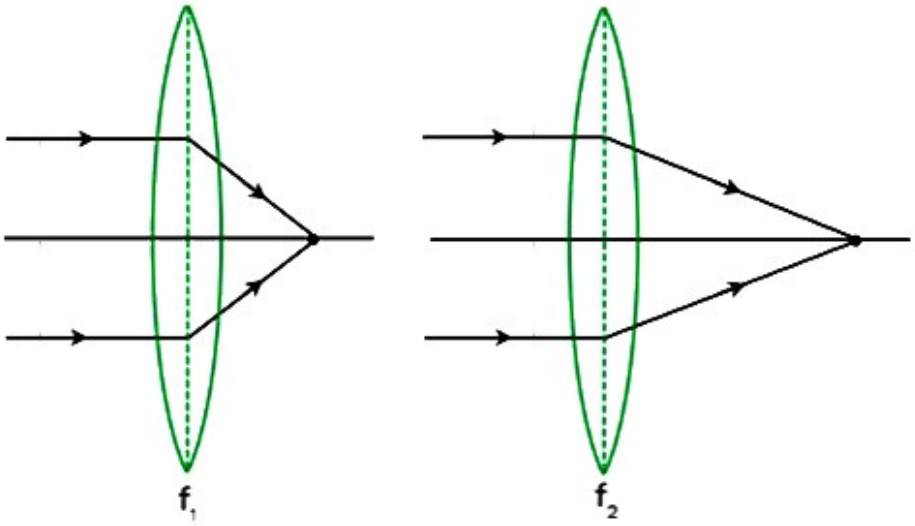
Substituting, u = −30 cm, v = +60 cm, $(V_{object,lens})_y$ = 5 cm/s, is Eq. (i),

$$\left(V_{image,lens}\right)_y = \left(\frac{+60}{-30}\right) \times 5 = -10\,cm/s$$

The image will be moving downwards with a velocity of 10 cm/s.

POWER OF A LENS

The power of a lens P is actually the measure of its ability to deviate the incident rays towards the axis. The greater the curve of the two surfaces (i.e., the shorter the focal length f), the greater is the lens action. The shorter the focal length of a lens, the more it converges or diverges the light, as shown in figure.



The power of a lens placed in the air is actually the reciprocal of the focal length of the lens in meter and is given by.

$$P = \frac{1}{f(\text{in metre})} = \frac{100}{f(\text{in cm})}$$

SI unit of power is dioptre (D)

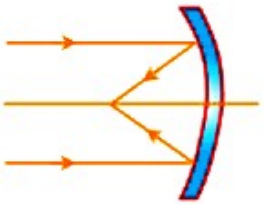
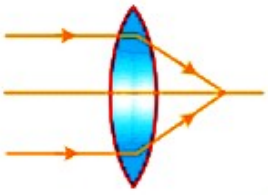
As per sign convention, the optical power of a converging lens (or convex lens with focal length positive) is positive.

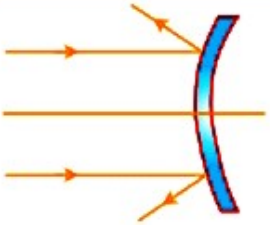
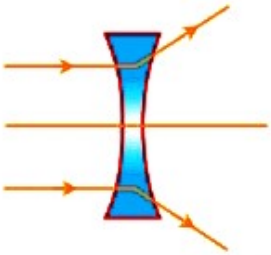
The optical power of a diverging lens (or concave lens with focal length negative) is taken negative.

Also we must note that for a mirror, power is defined as

$$P = \frac{1}{f(\text{in metre})} = \frac{100}{f(\text{in cm})}$$

Thus a convex lens and concave mirror have a converging nature, and hence they have positive power, whereas the concave lens and convex mirror have diverging nature and hence have negative power.

Nature of Lens Mirror	Focal Length (f)	Power $P_{lens} = \frac{1}{f}$, $P_{mirror} = -\frac{1}{f}$	Converging/ Diverging	Ray Diagram
Concave mirror	–ve	+ve	Converging	
Convex lens	+ve	+ve	Converging	

Nature of Lens Mirror	Focal Length (f)	Power $P_{lens} = \frac{1}{f}$, $P_{mirror} = -\frac{1}{f}$	Converging/ Diverging	Ray Diagram
Convex mirror	+ve	–ve	Diverging	
Concave Lens	–ve	–ve	Diverging	

Q **A convergent lens of power 6 D is combined with a diverging lens of –2 D. What is the optical power and focal length of the combination.**

Sol: Here $P_1 = 6$ D, $P_2 = -2$ D
Power of the combination is given by

$$P = P_1 + P_2 = 6 - 2 = 4D$$

Since $f = \frac{1}{P}$ $\Rightarrow$ $f = \frac{1}{4} = 0.25\,m = 25\,cm$.

Q **From a convex lens that has a focal length of 20 centimeters, an object of height 4 cm is kept at a distance of 10 cm. Find the nature, size, and position of the image formed. Also, calculate the optical power of the lens.**

Sol: Here, u = 10 c.m. (here we assumed that object is placed to the left side of optical centre.)
f = +20 cm (positive for a converging lens)

h_1 = +4 cm (since object kept above the principal axis)
Using the lens formula, we get

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{1}{v} - \frac{1}{-10} = \frac{1}{20}$$

$$\Rightarrow \quad v = -20 \text{ cm.}$$

The negative sign indicates that the image lies to the same side as the object is placed. Hence, the image is virtual. The linear magnification can be written as,

$$m = \frac{h_2}{h_1} = \frac{v}{u}$$

So, size of the image is

$$h_2 = h_1\left(\frac{v}{u}\right) = 4 \times \frac{-20}{-10} = 8\,\text{cm}$$

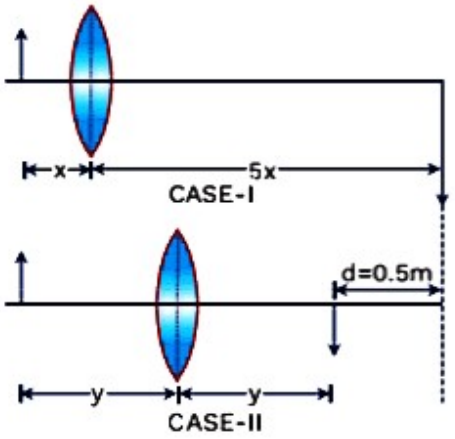
The sign came out to be positive that meant the image was erect and virtual.
Since the power of the lens is given by

$$P = \frac{1}{f(\text{in m})} = \frac{1}{+0.2} = +5\,D$$

Q **Given that a converging lens forms a magnified image of an object, the magnification is 5. Now the screen is moved towards the object by a distance of 0.5 meters, and the lens is shifted so that the image has the same size as the object. Calculate the lens power and also find the initial distance between the object and the screen.**

Sol: **Case 1:** Magnification is five times..
Hence, $|v| = 5|u|$

Case 2: Magnification= 1.
Hence, $|v| = |u|$



From the two figures, we get

$6x = 2y + d$

$\Rightarrow \quad 6x - 2y = 0.5$ ...(1)

Using the lens formula for both the cases, we get for

Case-I,

$$\frac{1}{5x} - \frac{1}{-x} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{6}{5x} = \frac{1}{f}$$ ...(2)

Case-II,

$$\frac{1}{y} - \frac{1}{-y} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{2}{y} = \frac{1}{f}$$ ...(3)

Solving these three equations, we get

$x = 0.1875$ m and $f = 0.15625$ m

Therefore, the initial distance between the object and the screen is

$6x = 1.125$ m

Power of the lens $P = \frac{P}{f} = \frac{1}{0.15625} D = 6.4 D$.

LENSES IN CONTACT

If two or more lenses of focal lengths f_1, f_2.... Are placed in contact, then their equivalent focal length f is given by

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + ... = \sum_{i=1}^{n} \frac{1}{f_i}$$

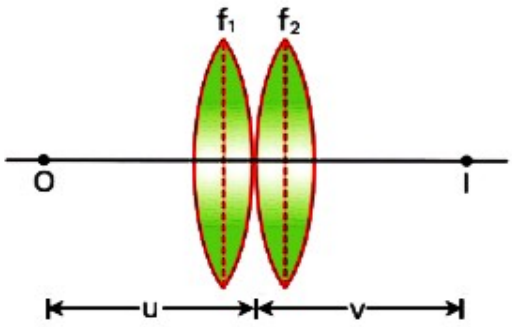
Where f_1, f_2.... are to be substituted with proper signs attached.
The power of combination

$$P = P_1 + P_2 + ...P_n = \sum_{i=1}^{n} P_i$$

Here too, P_1, P_2..... Are to be substituted with proper signs attached.
The magnification of the combination is

$$M = m_1 \times m_2 \times ... = \prod_{i=1}^{n} m$$

In many optical instruments, the combination of lenses in contact is used so as to improve the performance of the instrument.



Consider two lenses of focal lengths, f_1 and f_2, kept in contact. Let a point object O be placed at a distance u from the combination. The first image (say I_1) after refraction from the first lens is formed at a distance v_1 (whatever may be the sign of v_1) from the combination. This image I_1 acts as an object for the second lens and let v be the distance of the final image from the combination. Applying the lens formula $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$, we get

For the first lens, $\frac{1}{v_1}-\frac{1}{u}=\frac{1}{f_1}$ ...(1)

and for the second lens, $\frac{1}{v}-\frac{1}{v_1}=\frac{1}{f_2}$ ...(2)

Adding equations, (1) and (2), we get

$$\frac{1}{v}-\frac{1}{u}=\frac{1}{f_1}+\frac{1}{f_2}=\frac{1}{f} \text{ (say)}$$

where, F is the equivalent focal length of the combination.
Thus,

$$\boxed{\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}}$$

where, f is the equivalent focal length of the combination.

TWO THIN LENSES SEPARATED BY A DISTANCE

Suppose there are two lenses of focal lengths f_1, f_2, are placed at a distance x apart, then equivalent focal length of combination is

$$\frac{1}{F}=\frac{1}{f_1}+\frac{1}{f_2}-\frac{x}{f_1f_2}$$

or Power for the combination is

$$P = P_1 + P_2 - xP_1P_2$$

The net magnification of the combination is still remains

$$m = m_1 \times m_2$$

Q **Two convex lenses of focal lengths 30 cm and 70 cm. The refractive index of the lens is 1.5. They are held in contact coaxially by a rubber band around their edges. A liquid is introduced in the space between the lenses filling it completely, and the refractive index of the material is 1.3. Calculate the position of the image of a luminous object which is placed at a distance of 90 cm from it, on the axis of the combination lens.**

Sol: According to Lens Maker's Formula, we have

$$\frac{1}{30} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{-R_1}\right)$$

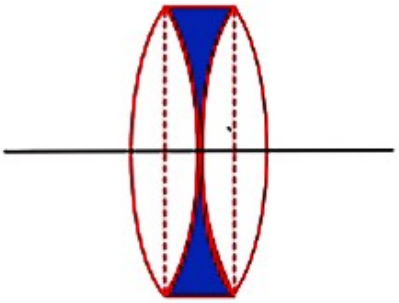
$\Rightarrow$ $R_1 = 30$ cm

Similarly, radius of curvature of the second lens is 70 cm.

Since

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} \qquad \ldots(1)$$

Here, $f_1 = 30$ cm, $f_2 = 70$ cm



Now, f_3 is calculated again using the Lens Maker's Formula, so we get

$$\frac{1}{f_3} = (1.3 - 1)\left(\frac{1}{-30} - \frac{1}{70}\right) \quad \Rightarrow \quad f_3 = -70 \text{ cm}$$

$\Rightarrow$ $F = 30$ cm {from equation (1)}

According to the Lens formula, applied on the combination of lenses, we have

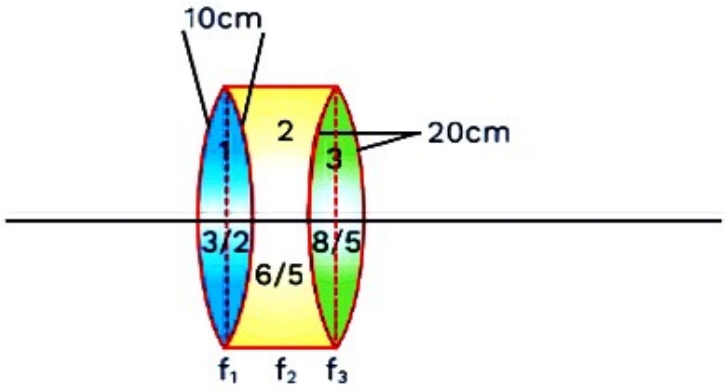
$$\frac{1}{v} - \frac{1}{u} = \frac{1}{F}$$

$$\Rightarrow \quad \frac{1}{v} - \frac{1}{(-90)} = \frac{1}{30}$$

$\Rightarrow$ $v = 45$ cm

Ans. The distance of the image is 45 cm from the lens.

Q **Find the focal length of the equivalent system shown in the figure.**



Sol: Here radius of curvatures of different surfaces and refractive index is given; we can calculate focal lengths of lenses by using

$$\frac{1}{f_1} = \left(\frac{3}{2} - 1\right)\left(\frac{1}{10} + \frac{1}{10}\right) = \frac{1}{2} \times \frac{2}{10} = \frac{1}{10}$$

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

$$\frac{1}{f_2} = \left(\frac{6}{5} - 1\right)\left(\frac{-1}{10} - \frac{1}{20}\right) = \frac{1}{5} \times \left(\frac{-30}{10 \times 20}\right) = \frac{-3}{100}$$

$$\frac{1}{f_3} = \left(\frac{8}{5} - 1\right)\left(\frac{1}{20} + \frac{1}{20}\right) = \frac{3}{50}$$

Hence, equivalent focal length

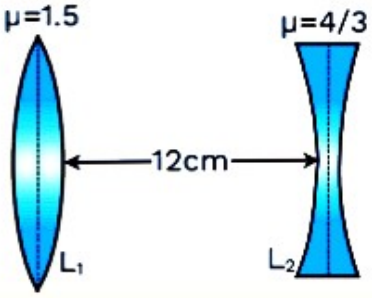
$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} = \frac{1}{10} + \frac{-3}{100} + \frac{3}{50}$$

$\Rightarrow$ f = 100/13 cm.

Problem Based on the Equivalent Focal Length of Thin Lenses Separated By a Distance From Each Other

Consider two lenses placed a distance d apart on the same optical axis. Here the object is placed at infinity, and the combination may be replaced by an equivalent single thin lens. This lens should converge the bundle of rays from infinity to the same point as the system of lenses and also do so with the same angle of convergence.

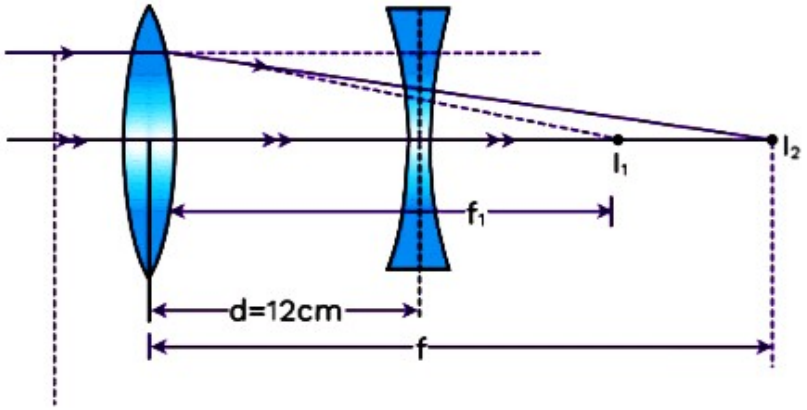
Q **A bi-convex lens L_1 of radii of curvature R_1 = 20 cm and R_2 = 30 cm and RI = μ = 1.5 is kept at a separation of 12 cm from a bi-concave lens L_2 whose radii of curvature R'_1 = 12 cm, R'_2 = 24 cm and RI = μ' = 4/3. Find the focal length of the effective lens.**



Sol: The focal length of the convex lens is given as

$$\frac{1}{f_1} = (1.5 - 1)\left(\frac{1}{20} + \frac{1}{30}\right)$$

or f_1 = 24 cm



The focal length of the concave lens is given as

$$\frac{1}{f_2} = -\left(\frac{4}{3} - 1\right)\left(\frac{1}{12} + \frac{1}{24}\right)$$

f_2 = −24 cm

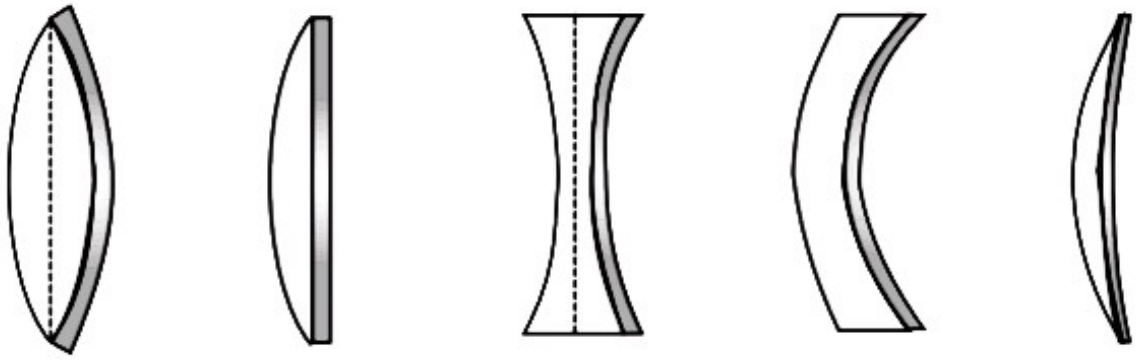
The effective focal length is given as

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2} = \frac{1}{24} + \frac{1}{-24} - \frac{12}{(24)(-24)}$$

or f = 48 cm.

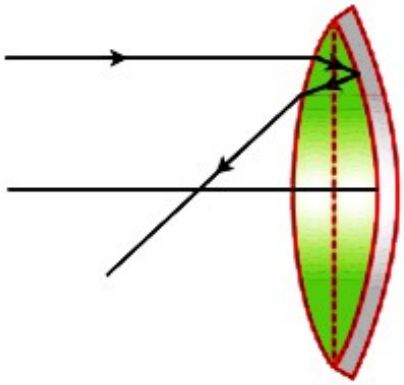
Silvering of lenses:

If we silvered one face of the lens, then it acts like a combination of mirror and lens.



Lenses with one face silvered act like lens-mirror combination

From the ray diagram, you can see that when the ray incident on the lens gets refracted after that, it is reflected from the silvered part, and during the return journey again, it gets refracted.



Since rays return to the same side of the incident ray and this happens in the mirror, that means the combination acts like a mirror whose effective power is given by

$$P_{net} = 2P_l + P_m$$

where P_l is the power of the lens and P_m is the power of the mirror.
Since for a mirror we have

$$P_m = -\frac{1}{f_m}$$

and for a lens, we have

$$P_l = \frac{1}{f_l} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

So, the combination acts like a mirror having a net focal length given by

$$F_{net} = \frac{1}{P_{net}}$$

$$P = -\frac{1}{F} = \frac{2}{f_l} - \frac{1}{f_m}$$

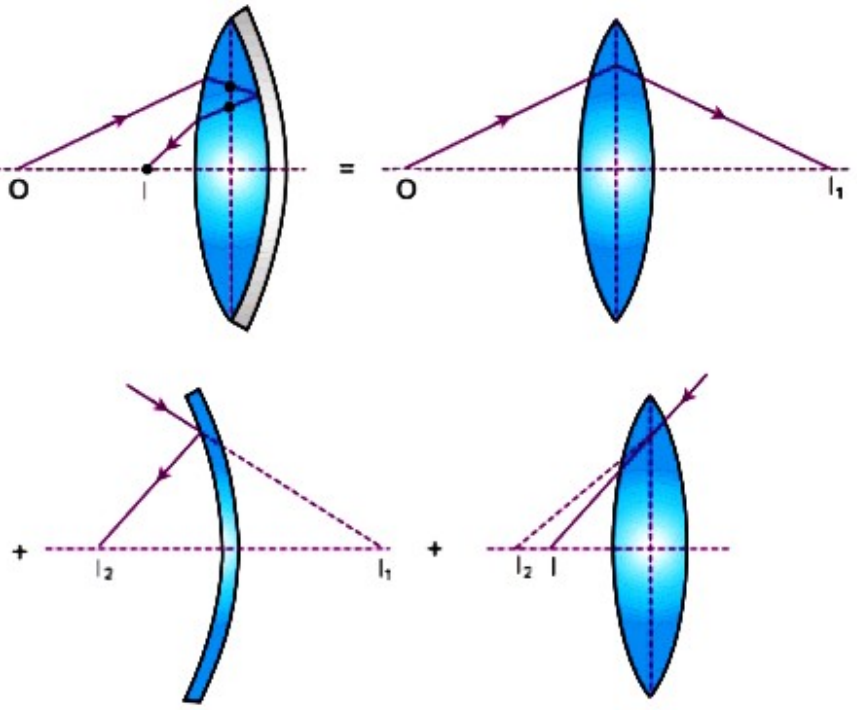
where f_l is focal length of lens and f_m is focal length of spherical mirror formed due to silvering of surface.

To have a fundamental understanding of this, we can understand the silvering of lenses using the following arguments.

(a) Light from object O passes through the lens to form image I_1.

(b) The image I_1 acts as a virtual object for the curved mirror to form image I_2.

(c) The image I_2 acts as a virtual object for the lens and the final image formed at I.



The silvered lens acts like a mirror with equivalent focal length F, given by

$$\frac{1}{F} = \frac{1}{f_l} - \frac{1}{f_m} + \frac{1}{f_l} = \frac{2}{f_l} - \frac{1}{f_m}$$

Where f_l is the focal length of the lens and f_m is the focal length of spherical mirror formed due to silvering of surface.

SIGN CONVENTION

While using the above formula, we make use of the following sign conventions.

(a) f is positive for converging (convex) lens and concave mirror.

(b) f is negative for diverging (concave) lens and convex mirror.

IMPORTANT POINTS

(a) Lenses must be thin

(b) Equivalent system at as a mirror

(c) Lenses must be in contact

(d) Put f_m, f_l with proper sign

(e) Equivalent focal length (f_{ec}) comes out with sign

(f) F_{ec} →
- +ve (Concave - mirror)
- –ve (Convex - mirror)
- ∞ (Plane - mirror)

Q **In plano-convex lens, the radius of curvature of the convex face is 12 cm. the refractive index is 1.5.**

(a) Calculate the focal length of a given lens. Now the lens is silvered at the plane side.

(b) Find out the distance from the lens at which parallel rays incident on the convex face converge.

(c) Sketch the ray diagram to locate the image when a point object is placed on the axis 20 cm from the lens.

(d) Calculate the image distance when the object is placed as in (c).

Sol: (a) As for a lens, by lens maker's formula,

$$\frac{1}{f} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$$

Here $\mu = 1.5$; $R_1 = 12$ cm; and $R_2 = \infty$

So, $\frac{1}{f} = (1.5 - 1)\left[\frac{1}{12} - \frac{1}{-\infty}\right]$ i.e., $f = 24$ cm

i.e., the lens is convergent with focal length 24 cm.

(b) As light after passing through the lens will be incident on the mirror which will reflect it back through the lens again.

so $P = P_L + P_M + P_L = 2P_L + P_M$

But $P_L = \frac{1}{f_L} = \frac{1}{0.24}$ and $P_M = -\frac{1}{\infty} = 0$

$\left[\text{as } f_M = \frac{R}{2} = \infty\right]$

So, $P = 2 \times \frac{1}{0.24} + 0 = \frac{1}{0.12} D$

The system is equivalent to a concave mirror of focal length F.

$$P = -\frac{1}{F}$$

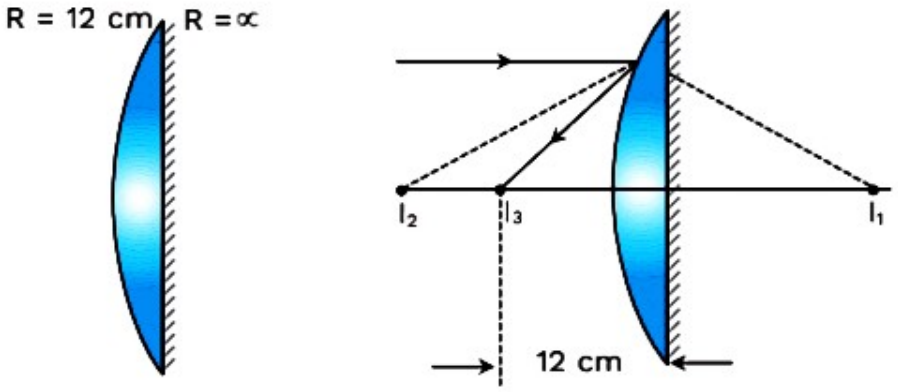
i.e., $F = -\frac{1}{P} = -0.12\,m = -12\,cm$

i.e., the system will behave as a concave mirror of focal length 12 cm. So, as for parallel incident rays u = $-\infty$, from mirror formula, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, we have

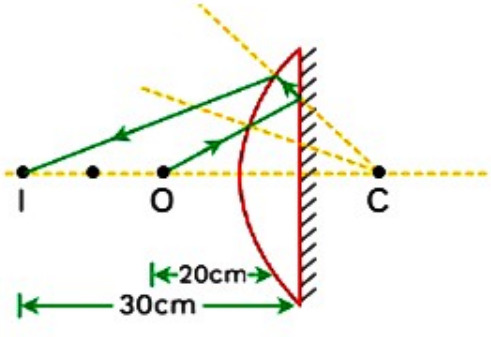
$$\frac{1}{v}+\frac{1}{-\infty}=\frac{1}{-12} \Rightarrow v = -12\,cm$$

i.e., parallel incident rays will focus at a distance of 12 cm in front of the lens, as shown in Figs. (a), (b) and (c). When an object is at 20 cm in front of the given silvered lens, which behaves as a concave mirror of focal length 12 cm, from mirror formula, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, we have

$$\frac{1}{v}+\frac{1}{-20}=\frac{1}{-12} \Rightarrow v = -30\,cm$$



(a) (b)

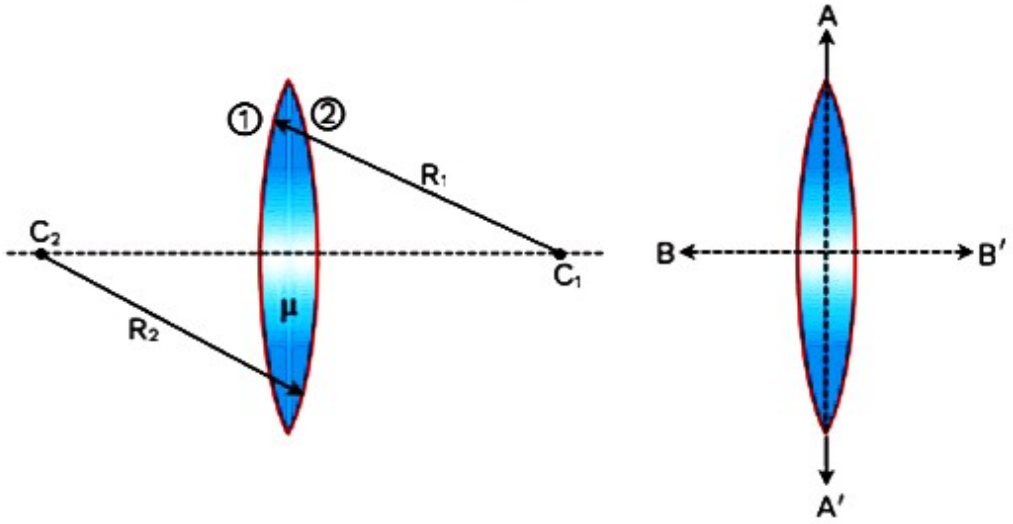


(c)

i.e., the silvered lens will form an image at a distance of 30 cm in front of a lens-mirror combination.

CUTTING THE LENS

Consider a biconvex lens of focal length f as shown in the figure.



We can relate its focal length with its radius of curvature as

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

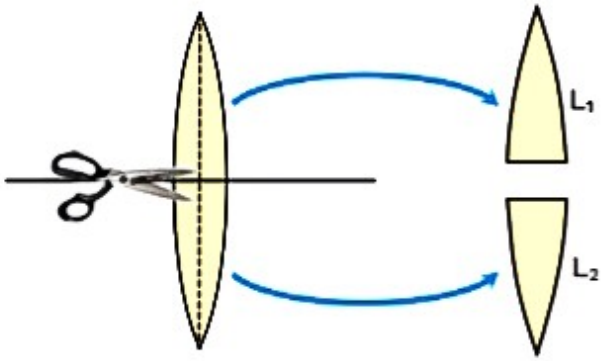
In this case $R_1 = +R$ and $R_2 = -R$

$$\frac{1}{f} = (\mu - 1)\left(\frac{1}{+R} - \frac{1}{(-R)}\right) = \frac{2(\mu - 1)}{R} \Rightarrow f = \frac{R}{2(\mu - 1)}$$

Now let us discuss how the lens will behave if it is cut in half along axis AA′ or along axis BB′? Will it behave as a lens? If yes, what is its new focal length?

Case -1 Lens cut along the principal axis (along BB′)

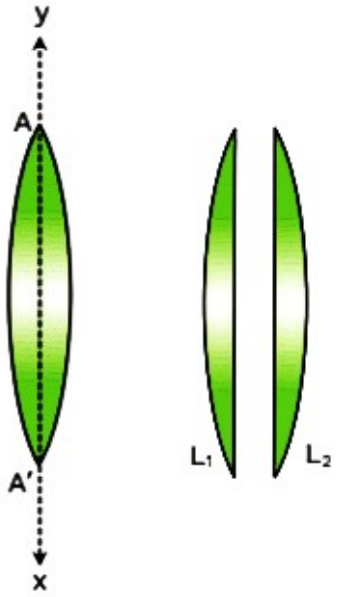
Consider a lens as shown in the figure. We cut the lens along the axis BB′. Now, we have two lenses, both of which are shown in the figure. Will the focal length of these two lenses change?



The answer is no, as the radii of curvature of the two surfaces of the lens still remain the same. The only intensity of the image decreases. If the lens is cut at the center, we can say that the image intensity will reduce by fifty percent.

Case -2 Lens cut perpendicular to the principal axis (along AA')

Now let us cut the lens along the perpendicular bisector AA'. We once again have two lenses, as shown in the figure.



Now we are going to calculate the focal length of half part of the lense.

For part L_1 $R_1 = R$ and $R_2 = \infty$

$$\frac{1}{f_{L_1}} = (\mu - 1)\left(\frac{1}{+R} - \frac{1}{\infty}\right) = \frac{(\mu - 1)}{R} \Rightarrow f_{L_1} = \frac{R}{(\mu - 1)} = 2f_0$$

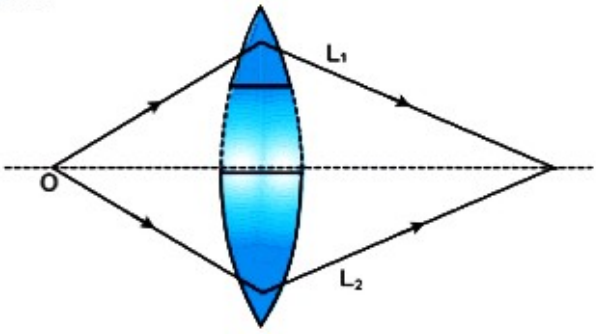
For part L_2 $R_1 = \infty$ and $R_2 = -R$

$$\frac{1}{f_{L_2}} = (\mu - 1)\left(\frac{1}{(\infty)} - \frac{1}{(-R)}\right) = \frac{(\mu - 1)}{R} \Rightarrow f_{L_2} = \frac{R}{(\mu - 1)} = 2f_0$$

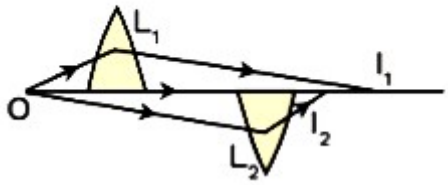
Therefore, the focal length is doubled.

IMPORTANT POINTS

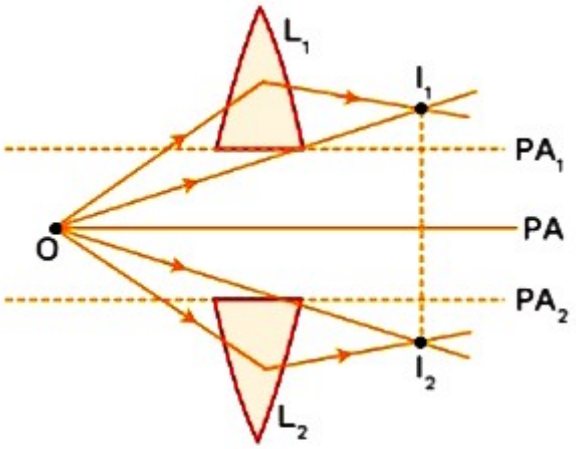
- If we cut and remove a portion of a lens, the principal axis (PA) is not disturbed. Hence the remaining two lenses behave like the original lens and form the same image of the object for the paraxial beam.



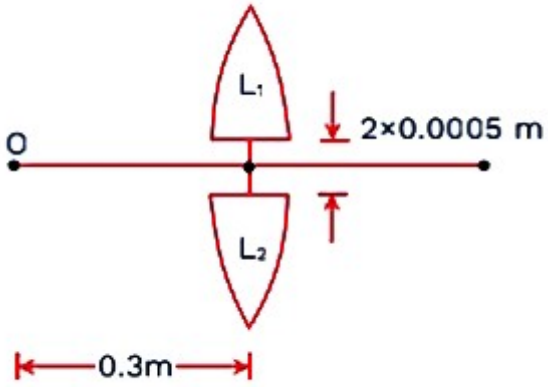
- If we cut the lens along (or parallel to) PA and shift them along with the PA, their PA does not change. Then, generally, two images will be formed on the PA for a single point object O placed on the PA.



- If we shift the cut lenses L_1 and L_2 perpendicular to the original PA, their principal axis will be shifted parallel to the original PA. Then I_1 and I_2 will be formed but not on the original PA.



Q **Suppose a convex lens of focal length 0.2 m and an object which is placed 0.3 m from the lens. Now the lens is cut into two halves and each of which is displaced by 0.0005 cm. Find the position of the image. Calculate the number of images formed and the distance between them if more than one image is formed.**



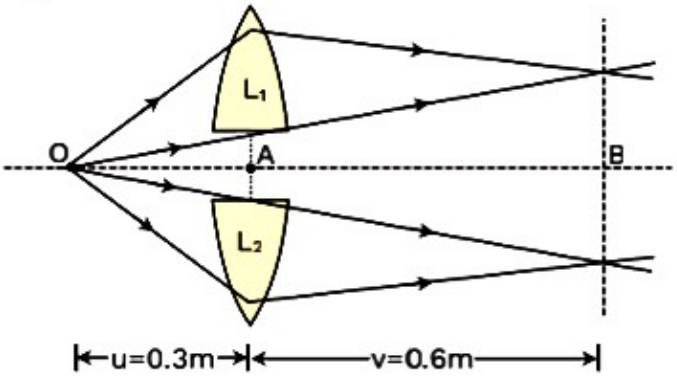
Sol: Each part will work as a separate lens and will form its own image. For any part, we have u = −0.3 m, f = +0.2 m.

Therefore, from lens formula, $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$, we have

$$\frac{1}{v}-\frac{1}{-0.3}=\frac{1}{0.2}$$

$\Rightarrow$ v = 0.6 m

So, each part forms a real image of the point object O at 0.6 m from the lens, as shown in the figure.



Since the triangles OL_1L_2 and OI_1I_2 are similar. So, we have

$$\frac{I_1I_2}{L_1L_2}=\frac{OB}{OA}=\frac{u+v}{u}$$

$$\Rightarrow \quad \frac{I_1I_2}{L_1L_2}=\frac{0.3+0.6}{0.3}=\frac{0.9}{0.3}=3$$

$\Rightarrow$ $I_1I_2 = 3(L_1L_2) = 3(2\times0.0005) = 0.003$ m.

CONCEPT OF IMAGE FORMING AT OBJECT ITSELF (CONCEPT OF AUTOCOLLIMATION)

In problems in optics, we will usually have a train optical element with the stipulation that the image is formed on the object itself. In such cases, there will have to be a mirror at the end of the optical train and the rays must be incident normally on the mirror in order to retrace their paths.

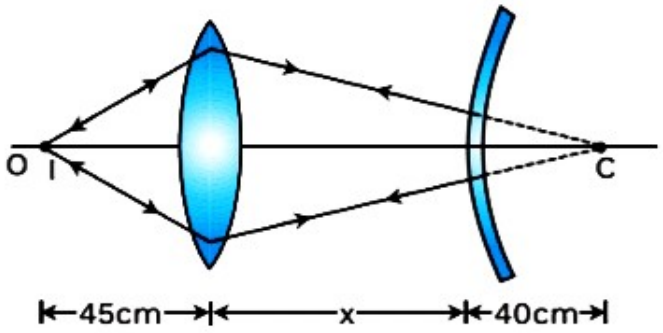
Q **Given that a converging lens of focal length 30 cm and an object is placed at a distance of 45 cm from the converging lens. Suppose a mirror of radius of curvature 40 cm is to be placed on the other side of the lens. Due to this, the object coincides with its image (autocollimation concept). Find the position of the mirror if it is**

(a) convex?

(b) concave?

Sol: The object and image will coincide only if the light ray retraces its path, and it will happen only when the ray strikes the mirror normally. In other words, the center of the curvature of the mirror and the rays incident on the mirror are collinear.

(a) The rays after refraction from lens must be directed towards the centre of curvature of mirror at C. If x is the separation, then for the lens
$u = -45$ cm, $v = x + 40$, $f = 30$ cm

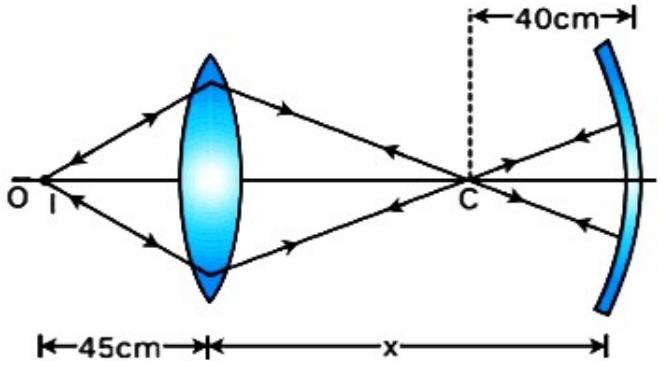


Using lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \quad \frac{1}{x+40} - \frac{1}{-45} = \frac{1}{30}$$

$$\Rightarrow \quad x = \frac{45(30)}{45-30} - 40 = 50\,\text{cm}$$

(b) In case of concave mirror, the refracted rays from lens meet at C, the centre of curvature (C) of the mirror.



Using lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ where $u = -45$ cm,
$v = x - 40$, $f = 30$ cm, we get

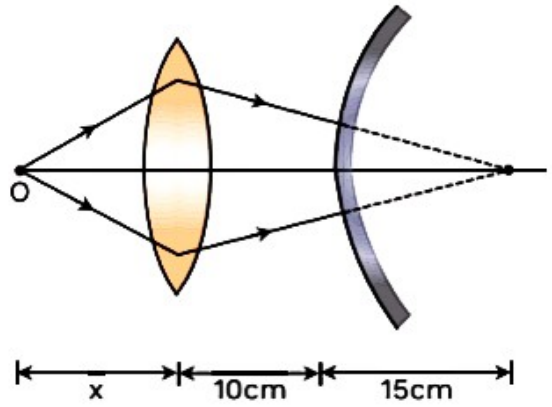
$$\frac{1}{x-40} - \frac{1}{-45} = \frac{1}{30}$$

$$\Rightarrow \quad x - 40 = \frac{45 \times 30}{45 - 30}$$

$$\Rightarrow \quad x = 90 + 40 = 130\,\text{cm}$$

Q **Given that a convex lens of focal length 20 cm and a mirror of radius of curvature 15cm. The lens is placed in front of a mirror at a distance of 10cm. Find the position of a point object to be placed in front of the lens so that images will form on the object itself**

Sol:



From the given condition, we can conclude that the combination behaves like a concave mirror. Let the distance of the object from the lens be x.

For the ray to retrace its path, it should be incident normally on the convex mirror, or in other words, and the rays should pass through the center of curvature of the mirror.

From the diagram, we see that for the lens

u = –x, f = +20 cm,

v = +10 + 15 = +25 cm

From the lens equation, we get $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

or $\frac{1}{25} - \frac{1}{-x} = \frac{1}{20}$

or x = 100 cm